Practice Problems
Calculus Physics II

Name: Key Date: _____

Chapter 59 – Imaging with Mirrors

1) The image you see in a plane mirror is

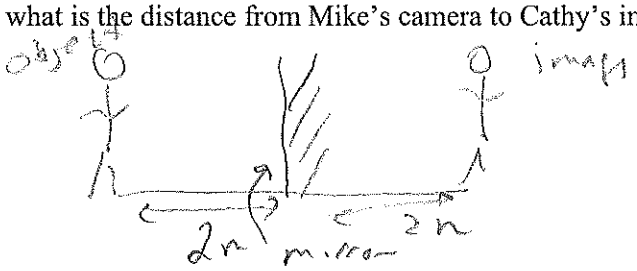
A) real. not at location it appears to be.
B) virtual.
C) both of these
D) neither of these

2) Is it possible to see a virtual image?

A) No, since the rays that seem to emanate from a virtual image do not in fact emanate from the image.
B) No, since virtual images do not really exist.
C) Yes, because the rays that appear to come from a virtual image can be focused by the eye just like those from an object.
D) Yes, but only by using an additional lens to form a real image before the light reaches the eye.

3) Mike and Cathy are standing together 2 m in front of a plane mirror, Mike points his camera into the mirror to take a picture of Cathy, what is the distance from Mike's camera to Cathy's image in the plane mirror?

A) 1 meter.
B) 3 meters.
C) 2 meters.
D) 4 meters.
E) none of the above



4) The shortest plane mirror in which you can see your entire image is

A) half your height.
B) about 3/4 your height.
C) about 1/3 your height.
D) equal to your height.
E) dependent on your distance from the mirror.



5) You may have seen ambulances on the street with the letters of the word AMBULANCE written on the front of them, in such a way as to appear correctly when viewed in your car's rear-view mirror. (See the figure.) How do the letters appear when you look directly at the ambulance (not through the mirror)?

A) a
B) b
C) c
D) d
E) e

a) ∀WBUL∀NƆE

b) AMBULANCE

c) ECNALUBMA

d) ƎƆNA⅃UBMA

e) ƎƆNA⅃UBMA

6) As Bob walk away from a plane mirror on a wall on a horizontal floor, the height of Bob's image

A) gets smaller.
B) may or may not get smaller, depending on where the observer is positioned.
C) is always a real image, no matter how far you are from the mirror.
D) changes from being a virtual image to a real image as you pass the focal point.
(E) is always the same size.

7) An object is located 3.8 m in front of a plane mirror. The image formed by the mirror appears to be

(A) 3.8 m behind the mirror's surface.
B) on the mirror's surface.
C) 1.9 m in front of the mirror.
D) 3.8 m in front of the mirror.
E) 1.9 m behind the mirror's surface.

8) Suppose you place an object in front of a concave mirror. Which of the following statements *must* be true?

A) The image of the object will always be smaller than the object.
B) No matter where you place the object, a real image of the object will be formed.
C) The image of the object will always be inverted.
(D) If you position the object between the mirror and the focal point of the mirror, its image must be upright and virtual.
E) No matter where you place the object, the image of the object will always be virtual and upright.

9) An object is placed in front of a spherical mirror if the image formed by the mirror is real, where is the imaged formed?

A) Opposite side of the mirror as the object.
(B) Same side of the mirror as the object.
C) Either opposite or same side as the object.
D) Can not be determined unless the type of mirror is known.

10) An object is placed in front of a spherical mirror if the image formed by the mirror is virtual, where is the imaged formed?

(A) Opposite side of the mirror as the object.
B) Same side of the mirror as the object.
C) Either opposite or same side as the object.
D) Can not be determined unless the type of mirror is known.

11) Can you tell the difference between a real and virtual image?

(A) No
B) Yes

12) If the object height is 10 cm and the image formed by the mirror has a height of 2 cm, what is the magnification of the object?

A) 5
B) 2
(C) 0.2

$$m = \frac{h_N}{h_O} = \frac{2cm}{10cm} = 0.2$$

13) If an object is placed 20 cm in front of a spherical mirror and the image is formed 5 cm in back of the mirror, what is the magnification of the mirror?

(A) + 0.25
B) - 0.25
C) + 4
D) - 4

$m = -\frac{d_i}{d_o} = -\frac{(-5cm)}{20cm}$ $= +0.25$

$d_i = -5cm$ ← Virtual

14) A negative magnification for a mirror means that

A) the image is inverted, and the mirror could be either concave or convex.
B) the image is upright, and the mirror is convex.
C) the image is inverted, and the mirror is convex.
(D) the image is inverted, and the mirror is concave.
E) the image is upright, and the mirror could be either concave or convex.

15) The focal length of a concave mirror has a magnitude of 20 cm. What is its radius of curvature?

A) 10 cm
(B) 40 cm
C) -40 cm
D) 20 cm
E) -20 cm

$f = +\frac{1}{2}R$ $\quad R = +2f = +40cm$

16) Single concave spherical mirrors produce images that

A) are always smaller than the actual object.
B) are always larger than the actual object.
C) are always the same size as the actual object.
(D) could be smaller than, larger than, or the same size as the actual object, depending on the placement of the object.
E) are always real.

17) Single convex spherical mirrors produce images that

(A) are always smaller than the actual object.
B) are always larger than the actual object.
C) are always the same size as the actual object.
D) could be larger than, smaller than, or the same size as the actual object, depending on the placement of the object.
E) are sometimes real.

18) A beam of light that is parallel to the principal axis strikes a concave mirror. What happens to the reflected beam of light?

A) It also is parallel to the principal axis.
B) It is perpendicular to the principal axis.
C) It passes through the center of curvature of the mirror.
(D) It passes through the focal point of the mirror.
E) It passes between the focal point and the center of curvature of the mirror.

19) A beam of light that is parallel to the principal axis strikes a convex mirror. What happens to the reflected beam of light?

A) It also is parallel to the principal axis.
B) It is perpendicular to the principal axis.
C) It appears to be coming from the focal point on the other side of the mirror.
D) It appears to be coming from the center of curvature of the mirror.
E) It appears to be coming from a point between the focal point and the center of curvature of the mirror.

20) An upright object is 50 cm from a concave mirror with a focal length of 30 cm. The character of the image is

A) real and upright.
B) real and inverted.
C) virtual and upright.
D) virtual and inverted.

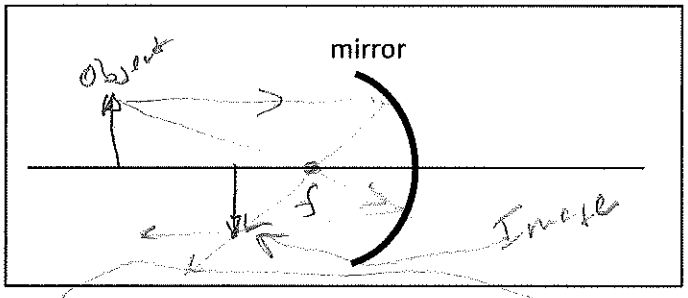
Out side focal length means real image

21) If the magnification of a mirror or lens is negative, it means that

A) the image is smaller than the object.
B) the image is inverted and smaller than the object.
C) the image is inverted.
D) the object is farther from the mirror (or lens) than the image.
E) the image is farther from the mirror (or lens) than the object.

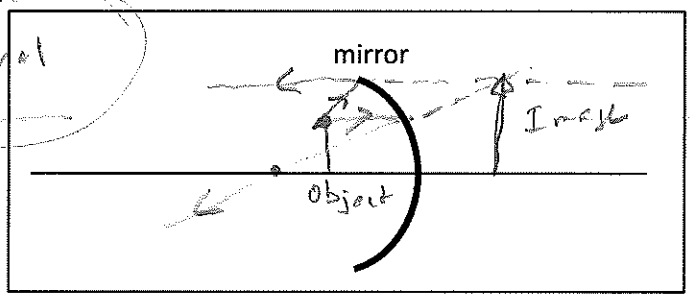
22) A light bulb is placed 60 cm in front of a concave mirror with a focal length of 20 cm. Draw the ray tracing diagram for this problem and solve for the image distance d_i and magnification? Is it real or virtual and is it upright or inverted?

$d_o = 60cm$
$f = 20cm$
$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$
$d_i = \frac{1}{(\frac{1}{f} - \frac{1}{d_o})} = +30cm$ real
$m = \frac{-d_i}{d_o} = -0.5$ inverted



23) Cathy places a penny 50 cm in front of a concave mirror with a focal length of 60 cm. Draw the ray tracing diagram for this problem and solve for the image distance d_i and magnification? Is it real or virtual and is it upright or inverted?

$d_o = 50cm$
$f = 60cm$
$d_i = \frac{1}{(\frac{1}{f} - \frac{1}{d_o})} = -300cm$ virtual
$m = \frac{-d_i}{d_o} = +6$ upright



24) Bob's face is 20 cm in front of a concave shaving mirror. If he observes his image to be twice as big and upright, what is the focal length of the mirror?

$m = +2$ upright

$d_o = 20\text{cm}$

$m = -\frac{d_i}{d_o}$

$d_i = -m d_o$

$d_i = -40\text{cm}$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$f = \frac{1}{\left(\frac{1}{d_o} + \frac{1}{d_i}\right)} = +40\text{cm}$ Converget mirror

25) Mike notes when he enters the store a convex spherical mirror is hanging from the ceiling which has a focal length of f = - 1 m, Mike (h_{Mike} = 1.85 m) he is 5 m from the mirror, what is the image distance of Mike and how tall does he appear to be in the mirror. Draw the ray tracing diagram for this problem and solve for the image distance di and magnification? Is it real or virtual and is it upright or inverted?

$f = -1\text{m}$

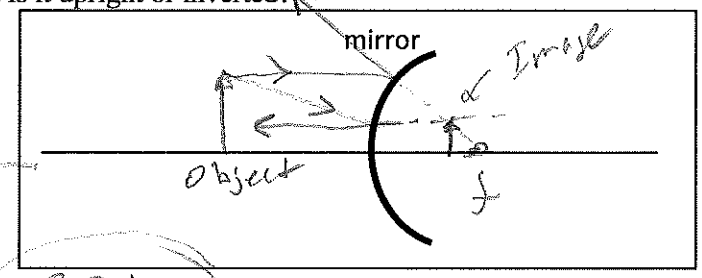
$d_o = 5\text{m}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -0.83\text{m}$ virtual

$m = -\frac{d_i}{d_o} = +0.17$ upright

$m = \frac{h_i}{h_o}$

$h_i = m h_o = 0.31\text{m}$



26) Cathy's face is 60 cm in front of a convex mirror and her image is upright, magnified by m = 0.25, what is the focal length of the convex lens?

$d_o = 60\text{cm}$

$m = +0.25$

$m = -\frac{d_i}{d_o}$

$d_i = -m d_o = -15\text{cm}$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$f = \frac{1}{\left(\frac{1}{d_i} + \frac{1}{d_o}\right)} = -20\text{cm}$ divegent mirror

27) A candle with a height of 5 cm is placed 11 cm in front of a concave mirror with a focal length of 10.5 cm, what is the height h_i of the image of the candle. Draw the ray tracing diagram for this problem and solve? Is it real or virtual and is it upright or inverted?

$h_o = 5\text{cm}$

$d_o = 11\text{cm}$

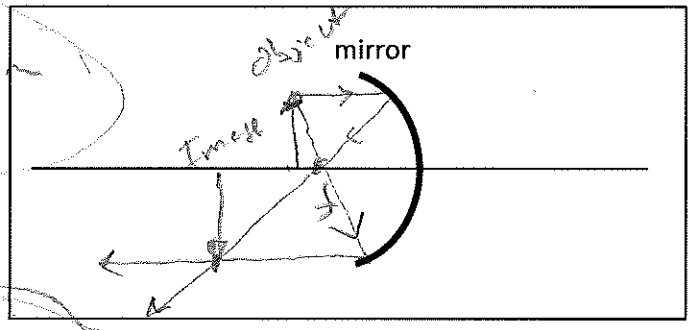
$f = +10.5\text{cm}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = +23\text{cm}$ real

$m = -\frac{d_i}{d_o} = -2.1$ Inverted

$m = \frac{h_i}{h_o}$

$h_i = m h_o = 10.5\text{cm}$



28) Cathy sees a car in her side view mirror (f = - 4 m) it is a car with a height of h_o = 2 m, if the car is d_o = 8 m behind Cathy's car mirror, what distance does it appear to Cathy (d_i) and what is the image height of the car (h_i)?

$f = -4\text{m}$

$h = 2\text{m}$

$d_o = 8\text{m}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -2.7\text{m}$

$m = -\frac{d_i}{d_o} = +0.33$

$m = \frac{h_i}{h_o}$

$h_i = m h_o = 0.67\text{m}$

29) A page in a book that is 8 cm tall is placed 5 cm in front of a spherical mirror, the image of the page has a height of 16 cm and is on the opposite side of the mirror as the book and is upright. What is the focal length of the mirror and what type of mirror is it?

$h_o = 8\text{cm}$

$d_o = 5\text{cm}$

$h_i = 16\text{cm}$

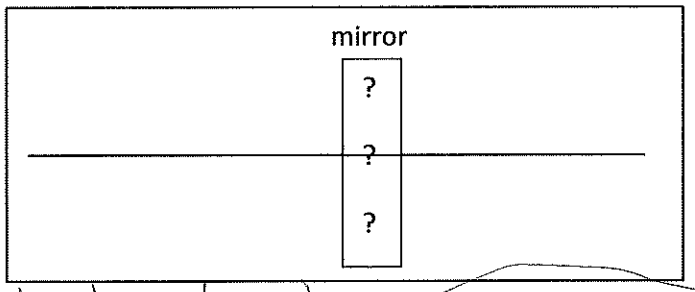
Virtual Image
↳ upright

$m = \frac{h_i}{h_o} = 2$

$m = -\frac{d_i}{d_o}$

$d_i = -m d_o$

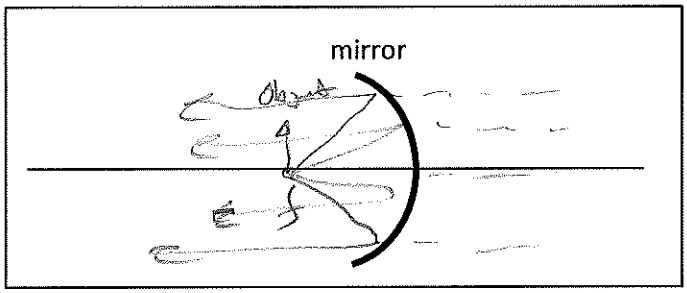
$d_i = -10\text{cm}$ (must be negative (virtual))



$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$f = \frac{1}{\left(\frac{1}{d_i} + \frac{1}{d_o}\right)} =$ converging mirror $+10\text{cm}$

30) A light bulb is placed 50 cm in front of a concave mirror with a focal length of 50 cm. Draw the ray tracing diagram for this problem and solve for the image distance d_i and magnification? Is it real or virtual and is it upright or inverted?



$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$d_o = f$

$d_i = \frac{1}{\frac{1}{f} - \frac{1}{d_o}} =$ No Image undefined

$d_i = \frac{1}{0}$

KEY

1) B
2) C
3) D
4) A
5) E
6) E
7) A
8) D
9) B
10) A
11) A
12) C
13) A
14) D
15) B
16) D
17) A
18) D
19) C
20) B
21) C
22) di = + 30 cm, m = - 0.5 image is inverted and real
23) di = - 300 cm, m = + 6 image is upright and virtual
24) di = - 40 cm, f = + 40 cm convergent mirror
25) hi = 0.31 m
26) di = - 15 cm and f = - 20 cm
27) hi = 105 cm
28) hi = 0.67 m
29) f = + 10 cm convergent mirror
30) no image is formed.